HR ANALYTICS PROJECT REPORT

Predicting Employee Salary & Attrition Risk

Using Linear Regression & Logistic Regression

Dataset: IBM HR Analytics Employee Attrition
Tool: Python (Jupyter Notebook) | April 2026

1. Executive Summary

This project applies machine learning techniques to IBM's HR Analytics dataset to answer two critical business questions that directly impact workforce strategy and cost management:

- How much salary should an employee earn based on their profile?
- Is this employee likely to leave the company?

By combining Linear Regression for salary prediction and Logistic Regression for attrition prediction on the same dataset, this project delivers a dual-model consulting framework that identifies underpaid, high-risk employees — enabling targeted retention interventions.

Key outcomes:

- Linear Regression achieved $R^2 = 0.8574$, explaining 85.7% of salary variance
- Logistic Regression achieved AUC-ROC = 0.783, a strong discrimination score
- 110 employees identified as Critical — both underpaid and high attrition risk
- Sales Representatives carry the highest combined risk at 72.8% average attrition

2. Problem Statement

2.1 Business Context

An organization is experiencing significant employee turnover and suspects that pay inequity may be a contributing factor. The HR leadership has commissioned a data analytics project to quantify both the compensation gap and the attrition risk at an individual employee level.

2.2 Business Questions

Model	Business Question	Target Variable	Output Type
Linear Regression	How much should this employee earn?	MonthlyIncome	Continuous (salary in $)
Logistic Regression	Will this employee leave?	Attrition (Yes/No)	Binary (0 or 1)

2.3 Why Both Models Together?

The consulting value emerges when both models are combined. An employee who is both underpaid (Linear Regression) and a flight risk (Logistic Regression) represents the highest priority retention case. This dual-model approach mirrors real-world advisory deliverables where data supports actionable recommendations.

3. Dataset Overview

3.1 Source

- Name: IBM HR Analytics Employee Attrition Dataset
- Source: Kaggle — publicly available
- File: WA_Fn-UseC_-HR-Employee-Attrition.csv

3.2 Dataset Dimensions

Attribute	Value
Total Rows (Employees)	1,470
Total Columns (Features)	35
Missing Values	0 (completely clean)
Target 1 — MonthlyIncome	Continuous, range: $1,009 to $19,999
Target 2 — Attrition	Binary: Yes (237, 16.1%) / No (1,233, 83.9%)

3.3 Key Features Used

Feature	Type	Description
JobLevel	Numeric	Seniority level (1-5) — strongest salary predictor
JobRole	Categorical	Employee's job title — 9 unique roles
TotalWorkingYears	Numeric	Total years of professional experience
OverTime	Binary	Whether employee works overtime — top attrition driver
JobSatisfaction	Ordinal (1-4)	Self-reported job satisfaction score
YearsSinceLastPromotion	Numeric	Years since last promotion
Department	Categorical	HR, Sales, Research & Development
MonthlyIncome	Numeric	Actual monthly salary (Linear target)
Attrition	Binary	Whether employee left (Logistic target)

4. Exploratory Data Analysis (EDA)

4.1 Key Findings

Salary Distribution

Monthly income is strongly right-skewed — the majority of employees earn between $1,000 and $5,000, with a long tail towards $20,000. This skewness required a log transformation before feeding the data into Linear Regression to normalize the distribution.

Class Imbalance in Attrition

The attrition target is heavily imbalanced — 83.9% of employees stayed (No) vs 16.1% who left (Yes). This required special handling in the Logistic Regression model using class_weight='balanced' to prevent the model from simply predicting 'No' for everyone.

Salary vs Attrition

Employees who left had a significantly lower median salary (~$3,000) compared to those who stayed (~$5,500). This was one of the first indicators that salary is strongly linked to attrition risk.

Attrition by Department

Department	Attrition Rate	Observation
Sales	~19%	Highest attrition — urgent intervention needed
Human Resources	~19%	Small dept but proportionally very risky
Research & Development	~14%	Lowest rate but largest headcount

Correlation Heatmap Findings

- JobLevel showed the strongest correlation cluster — confirming its role as the primary salary driver
- Most features had weak to moderate correlations — reducing multicollinearity concerns
- OverTime and JobSatisfaction showed meaningful signals for attrition

5. Data Preprocessing

5.1 Steps Taken

Step	Action	Reason
1	Dropped 4 columns	EmployeeCount, EmployeeNumber, StandardHours, Over18 — zero variance, no predictive value
2	Encoded Attrition	Mapped Yes=1, No=0 for Logistic Regression target
3	Label Encoded 2 columns	BusinessTravel and OverTime have natural order
4	One Hot Encoded 5 columns	Department, EducationField, Gender, JobRole, MaritalStatus — no natural order
5	Log transformed MonthlyIncome	Fixed right skew observed in salary distribution
6	StandardScaler applied	Required for Logistic Regression — equalizes feature scales
7	Stratified train/test split	80/20 split preserving attrition class ratio in both sets

5.2 Encoding Strategy

A deliberate encoding strategy was chosen to ensure model quality:

Encoding Type	Columns	Why
Label Encoding	BusinessTravel, OverTime	These have meaningful order: Non-Travel < Rarely < Frequently
One Hot Encoding	Department, Gender, JobRole, etc.	No natural order — prevents model from implying false ranking

5.3 Final Data Shape

- Original columns: 35
- After dropping 4 useless columns: 31
- After One Hot Encoding added new columns: 44
- After adding MonthlyIncome_log: 45 total
- Features used for modeling (X): 42 columns

6. Model 1 — Linear Regression

6.1 Objective

Predict the expected monthly salary of an employee based on their professional profile.

6.2 Model Performance

Metric	Value	Interpretation
R^2 Score	0.8574	Model explains 85.74% of salary variance — strong performance
RMSE	0.2368	Average prediction error of ~26% in actual salary terms
Training Size	1,176 employees	80% of 1,470
Testing Size	294 employees	20% of 1,470

6.3 Top 10 Salary Drivers

Feature	Coefficient	Impact on Salary
JobRole_Sales Representative	-0.415	Sales Reps earn ~34% LESS than baseline role
JobLevel	+0.360	Each level up adds ~43% to salary — strongest positive driver
JobRole_Laboratory Technician	-0.351	Lab Technicians earn ~30% less than baseline
JobRole_Research Scientist	-0.334	Research Scientists earn ~28% less
JobRole_Research Director	+0.189	Research Directors earn ~21% premium
JobRole_Manager	+0.188	Managers earn ~21% more than baseline
JobRole_Human Resources	-0.145	HR roles earn ~14% less
Department_R&D	+0.108	R&D employees get an ~11% salary premium
Department_Sales	+0.081	Sales dept earns ~8% more than HR dept
PerformanceRating	-0.047	Weakest driver — negligible salary impact

6.4 Key Insights

- JobLevel is the single most important positive driver — the company's pay is largely seniority-based
- JobRole matters more than Department — being a Sales Representative or Lab Technician significantly suppresses salary
- Performance Rating has almost no impact on salary — suggesting compensation is not merit-driven
- The Actual vs Predicted plot showed tight clustering around the diagonal, confirming genuine model accuracy
- The Residual Plot showed slight heteroscedasticity at lower salary ranges — model is less precise for junior employees

6.5 Consulting Headline

> *Job level and job role explain 85.7% of salary variance — the company's compensation structure is role-driven, not merit-driven. High performers in low-tier roles are systematically underpaid.*

7. Model 2 — Logistic Regression

7.1 Objective

Predict whether an employee is likely to leave the company (attrition = Yes) based on their professional and personal profile.

7.2 Model Performance

Metric	Value	Interpretation
Accuracy	73.47%	Misleading due to class imbalance — use AUC-ROC instead
AUC-ROC	0.783	Model correctly ranks leavers vs stayers 78.3% of the time — solid score
Recall (Attrition class)	0.60	Caught 60% of employees who actually left — useful for HR action
Precision (Attrition class)	0.32	When model flags risk, correct 32% of the time — some false alarms
Training attrition rate	16.16%	Stratify preserved class balance correctly
Testing attrition rate	15.99%	Matches training — no data leakage

7.3 Confusion Matrix Breakdown

	Predicted: No	Predicted: Yes
Actual: No (247 employees)	188 — Correctly identified as staying	59 — False alarms (flagged wrongly)
Actual: Yes (47 employees)	19 — Missed leavers (costly)	28 — Correctly caught leavers

Of 47 employees who actually left, the model correctly identified 28 (60%). While 59 false alarms were raised, the cost of a retention check is far lower than losing a valued employee.

7.4 Top 10 Attrition Drivers

Feature	Coefficient	Business Meaning
OverTime	+0.763	Biggest driver — burned-out employees are most likely to quit
JobRole_Laboratory Technician	+0.758	Lab Technicians face very high flight risk
TotalWorkingYears	-0.636	More experienced employees are LESS likely to leave — loyalty grows with tenure

Feature	Coefficient	Business Meaning
JobLevel	+0.588	Senior employees surprisingly have higher attrition — better external options
JobRole_Sales Representative	+0.563	Sales Reps are high flight risk
Department_Sales	+0.525	Entire Sales department is elevated risk
EducationField_Life Sciences	-0.495	Life Sciences graduates tend to stay longer
YearsSinceLastPromotion	+0.487	Longer without promotion = higher likelihood of leaving
EducationField_Medical	-0.460	Medical backgrounds show higher loyalty
NumCompaniesWorked	+0.452	Job-hoppers tend to continue job-hopping

7.5 ROC Curve Interpretation

- The ROC curve bulges strongly toward the top-left corner — indicating a genuinely useful model
- AUC = 0.783 is well above the 0.5 random baseline
- The early steep rise confirms high confidence predictions for the riskiest employees

7.6 Consulting Headline

> *Overtime is the single biggest predictor of attrition. Employees doing overtime in junior Sales and Lab Technician roles face a near-certain exit risk. An overtime policy review could be the highest-ROI retention intervention available.*

8. Combined Risk Analysis

8.1 Methodology

Both model outputs were merged into a single employee-level risk table combining predicted salary, actual salary, salary gap, and attrition probability. Each employee was assigned a risk category:

Risk Category	Criteria	Action Required
Critical (Red)	Attrition risk >= 60% AND underpaid	Immediate salary correction + retention meeting
High Risk (Orange)	Attrition risk >= 60% but fairly paid	Non-monetary intervention: growth, culture, overtime
Moderate (Yellow)	Underpaid but attrition risk < 60%	Proactive salary review before risk escalates
Low Risk (Green)	Fairly paid and low attrition risk	No action needed — stable workforce

8.2 Workforce Risk Summary

Metric	Value	Consulting Implication
Total Employees	1,470	Full workforce assessed
Critical Risk Employees	110 (7.5%)	Immediate intervention required
High Risk Employees	231 (15.7%)	Retention programs needed
Underpaid Employees	471 (32%)	Nearly 1 in 3 employees underpaid
Average Attrition Risk	36.1%	High baseline risk across workforce
Average Salary Gap	-$85/month	Widespread but small underpayment

8.3 Risk by Department

Department	Avg Attrition Risk	Critical Employees	Avg Salary Gap	Total Employees
Sales	45.1%	42	-$104	446
Human Resources	41.4%	6	-$353	63
Research & Development	31.6%	62	-$59	961

8.4 Risk by Job Role

Job Role	Avg Attrition Risk	Critical Count	Avg Salary Gap
Sales Representative	72.8%	16	-$108
Laboratory Technician	51.5%	35	-$170
Human Resources	47.1%	6	-$441
Sales Executive	40.7%	24	-$179
Research Scientist	31.9%	23	-$165
Manufacturing Director	23.8%	2	-$140
Healthcare Representative	21.3%	2	-$200

8.5 Top Critical Employee Profile

The highest-risk employees identified share this profile:

Attribute	Profile of Critical Employee
Department	Research & Development or Sales
Job Role	Laboratory Technician or Sales Representative
Job Level	1 (Junior)
Actual Salary	$1,118 to $2,340/month
Predicted Salary	$2,260 to $3,050/month
Salary Gap	$320 to $1,142 underpaid
Attrition Risk	94.7% to 99.1%

9. Recommendations

9.1 Immediate Actions (0-3 months)

- Conduct salary correction for 110 Critical employees — average gap of ~$750/month represents a small cost vs. rehiring expense of 50-200% of annual salary
- Review and reduce overtime policies in Sales and R&D departments — OverTime is the single biggest attrition driver
- Prioritize retention conversations for Sales Representatives — 72.8% average attrition risk is unsustainable

9.2 Medium-term Actions (3-6 months)

- Implement a promotion review cycle for employees with 2+ years without promotion — YearsSinceLastPromotion is a top-5 attrition driver
- Redesign compensation bands for Laboratory Technicians — both underpaid and high attrition risk
- Create career development pathways for junior Sales and R&D staff to reduce flight risk

9.3 Long-term Strategy (6-12 months)

- Shift compensation philosophy from role-based to merit-based — PerformanceRating currently has almost no salary impact
- Build a real-time attrition risk dashboard using this model to flag emerging cases
- Re-run the model every 6 months as workforce composition changes

10. Model Comparison Summary

Aspect	Linear Regression	Logistic Regression
Objective	Predict salary amount	Predict attrition probability
Target Variable	MonthlyIncome (continuous)	Attrition — 0 or 1 (binary)
Key Metric	$R^2 = 0.8574$	AUC-ROC = 0.783
Top Feature	JobLevel (+0.360)	OverTime (+0.763)
Evaluation Plots	Actual vs Predicted, Residual Plot	Confusion Matrix, ROC Curve
Special Handling	Log transform on target	Class weight balancing, StandardScaler
Business Output	Expected salary per employee	Attrition risk % per employee
Consulting Use	Identify underpaid employees	Identify flight-risk employees

11. Conclusion

This project successfully demonstrated the application of two complementary regression techniques to a real-world HR dataset. The combination of Linear and Logistic Regression on the same dataset created a dual-lens view of workforce risk that mirrors the kind of analysis a management consulting firm would deliver to an HR or COO client.

The key finding — that employees in Sales Representative and Laboratory Technician roles are simultaneously the most underpaid and the most likely to leave — directly answers the client's strategic question and provides a clear, data-backed basis for intervention.

Beyond the technical execution, this project demonstrates the ability to:

- Frame a business problem into a data science question
- Make deliberate, justified preprocessing decisions (encoding strategy, log transformation, class balancing)
- Interpret model outputs in business terms, not just statistical terms
- Synthesize two models into a single actionable consulting deliverable

Final Consulting Statement: Of 1,470 employees, 110 are in critical risk — underpaid and likely to leave. Immediate salary corrections in Sales Representative and Laboratory Technician roles, combined with an overtime policy review, represent the highest-ROI retention investment available to this organization.

12. Technical Appendix

12.1 Tools & Libraries

Library	Version	Purpose
Python	3.x	Core programming language
pandas	Latest	Data manipulation and analysis
numpy	Latest	Numerical operations and log transformation
scikit-learn	Latest	Linear Regression, Logistic Regression, preprocessing
matplotlib	Latest	Data visualization
seaborn	Latest	Statistical plots (heatmap, boxplot, countplot)
Jupyter Notebook	Latest	Interactive development environment

12.2 Project Steps

Step	Action	Output
1	Data loading & basic checks	Shape, dtypes, missing values, class distribution
2	Exploratory Data Analysis	5 visualization plots revealing key patterns
3	Preprocessing & encoding	Clean 45-column model-ready dataset
4	Linear Regression	R^2=0.8574, salary predictions, coefficient table
5	Logistic Regression	AUC=0.783, attrition probabilities, ROC curve
6	Combined Risk Table	Employee risk map with actionable segments

12.3 Key Formulas

Log Transformation applied to target variable:

MonthlyIncome_log = log(MonthlyIncome) [to normalize right skew]

Back-transformation for prediction output:

Predicted Salary = e^(predicted_log_income)

Attrition Risk Score:

Risk % = predict_proba(employee_features)[:, 1] * 100

Salary Gap:

Salary Gap = Predicted Salary - Actual Salary [positive = underpaid]